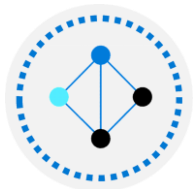


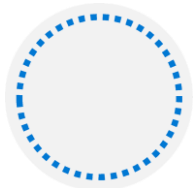
Training Course

Module 05: Network Traffic

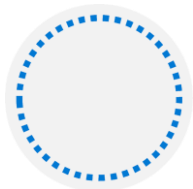




Lesson 01: Configure Network Routing and Endpoints

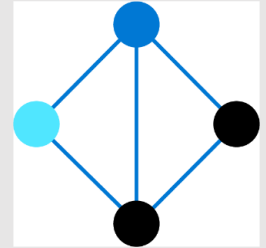


Lesson 02: Configure Azure Load Balancer



Lesson 03: Configure Application Gateway

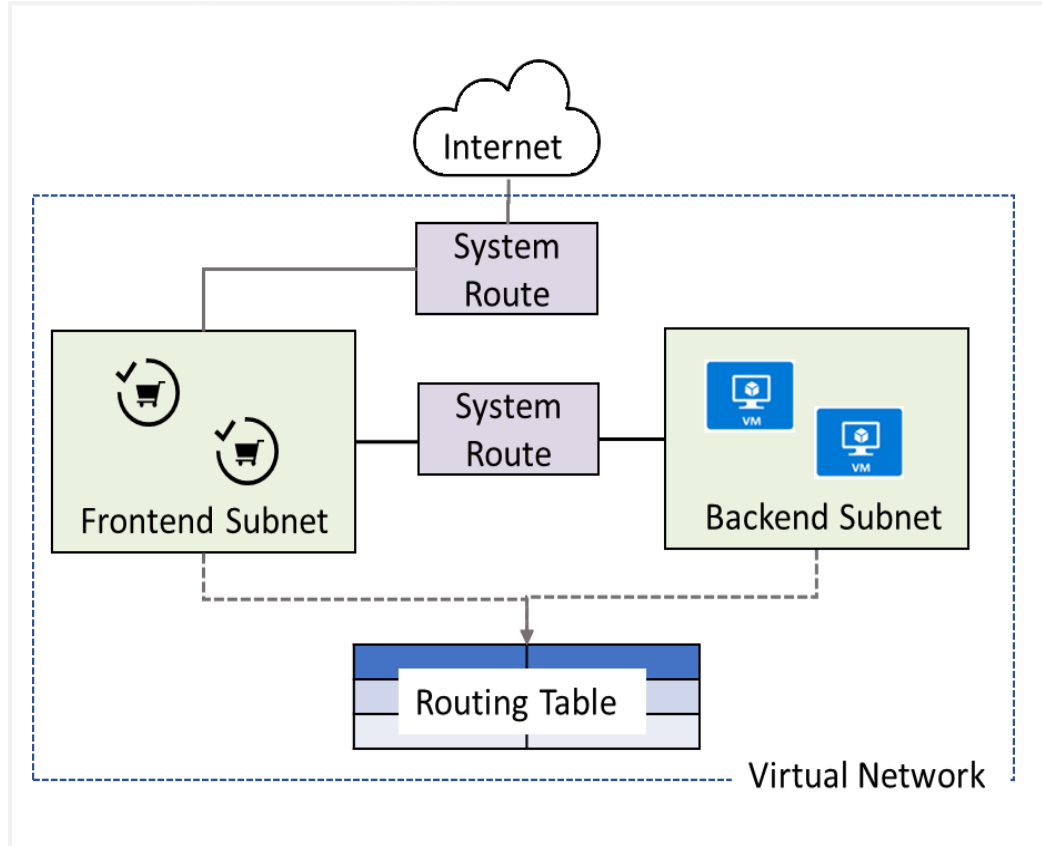
Lesson 01: Configure Network Routing and Endpoints



Review System Routes

System routes direct network traffic between virtual machines, on-premises networks, and the Internet:

- Traffic between VMs in the same subnet
- Between VMs in different subnets in the same virtual network
- Data flow from VMs to the Internet
- Communication between VMs using a VNet-to-VNet VPN
- Site-to-Site and ExpressRoute communication through the VPN gateway

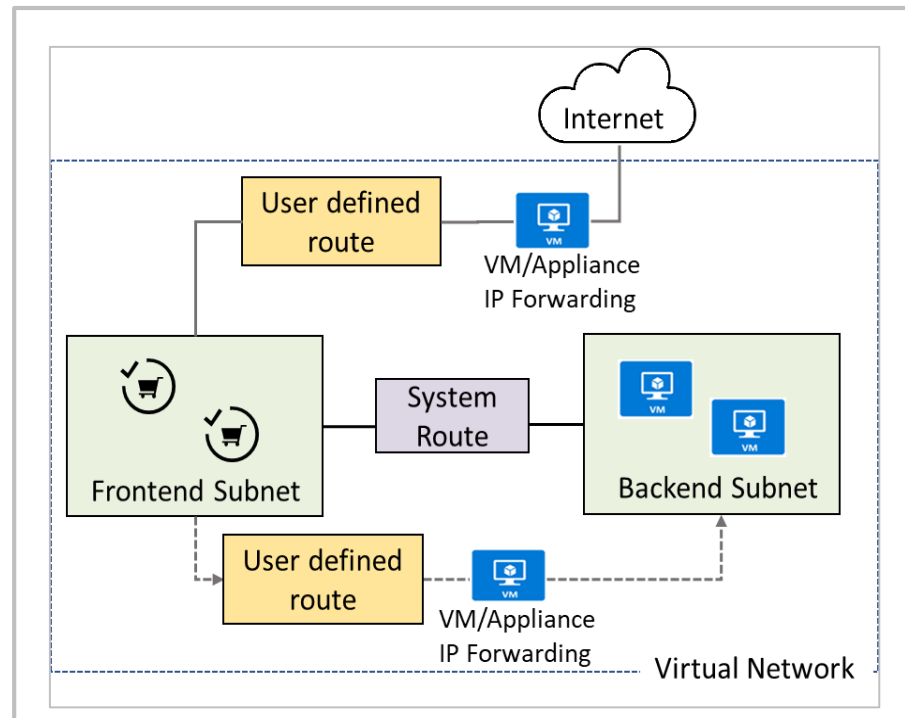


Identify User-Defined Routes

A route table contains a set of rules, called routes, that specifies how packets should be routed in a virtual network

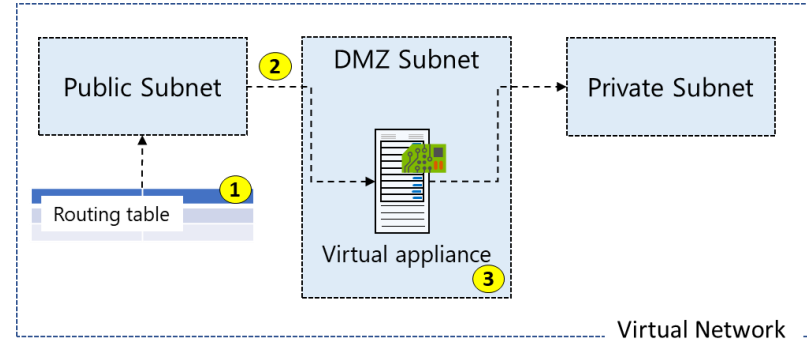
User-defined routes are custom routes that control network traffic by defining routes that specify the next hop of the traffic flow

The next hop can be a virtual network gateway, virtual network, internet, or virtual appliance



Examine a Routing Example

All traffic coming into the public subnet and headed for the private subnet must go through the virtual network appliance



Create route table 1

You can add routes to this table after it's created.

* Name
myRouteTablePublic ✓

* Subscription
Visual Studio Enterprise ✓

* Resource group
myRGWest ✓
[Create new](#)

* Location
(US) West US ✓

Virtual network gateway route propagation
Disabled **Enabled**

Create Automation options

Add route 2
myRouteTablePublic

Route name *
ToPrivateSubnet ✓

Address prefix * ⓘ
10.0.1.0/24 ✓

Next hop type ⓘ
Virtual network gateway ^

Virtual network gateway
Virtual network
Internet
Virtual appliance

None

Add subnet 3
VNet1

Name *
Public ✓

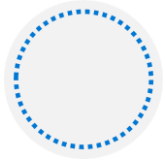
Address range (CIDR block) * ⓘ
10.0.1.0/24 ✓
10.0.1.0 - 10.0.1.255 (251 + 5 Azure reserved addresses)

NAT gateway ⓘ
None ✓
☐ Add IPv6 address space

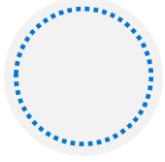
Network security group
None ✓

Route table
myRouteTablePublic ✓

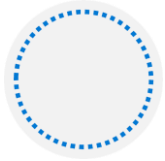
Demonstration – Custom Routing Tables



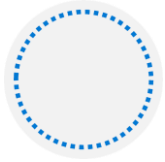
Create a route table



Add a route



Associate a route table to a subnet



Use PowerShell to view your routing information

Determine Service Endpoint Uses

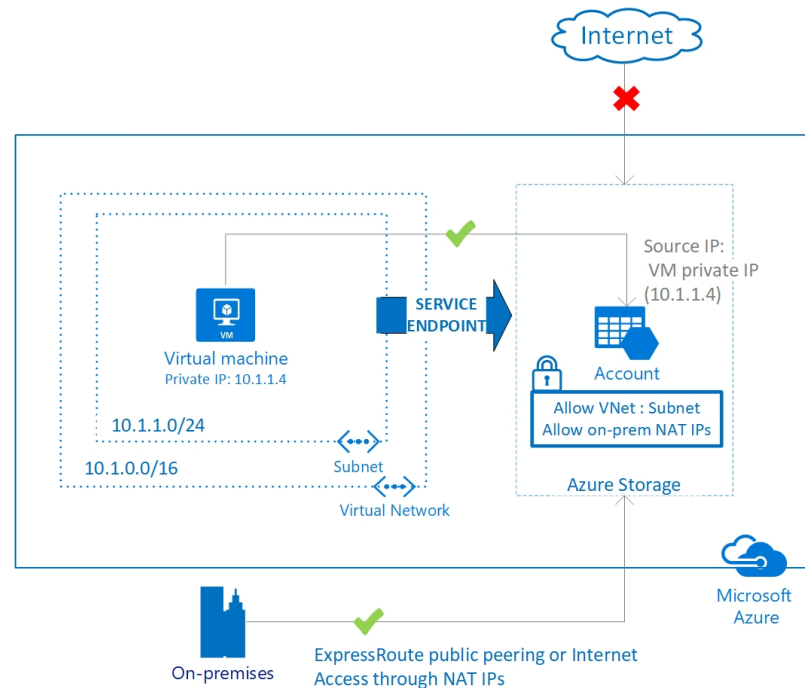
Endpoints limit network access to specific subnets and IP addresses

Improved security for your Azure service resources

Optimal routing for Azure service traffic from your virtual network

Endpoints use the Microsoft Azure backbone network

Simple to set up with less management overhead



Determine Service Endpoint Services

There are many types of service endpoints

Adding service endpoints can take up to 15 minutes to complete

Add service endpoints

Service *

Microsoft.Storage

Filter services

Microsoft.AzureActiveDirectory

Microsoft.AzureCosmosDB

Microsoft.CognitiveServices

Microsoft.ContainerRegistry

Microsoft.EventHub

Microsoft.KeyVault

Microsoft.ServiceBus

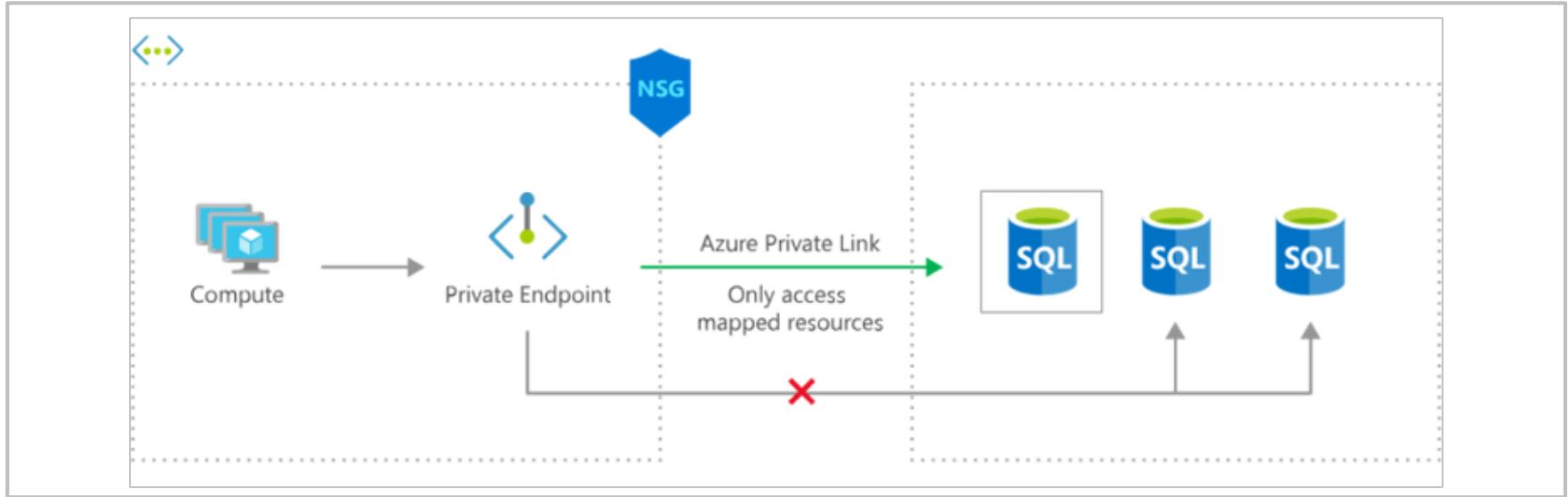
Microsoft.Sql

Microsoft.Storage

Microsoft.Web

Add

Identify Private Link Uses

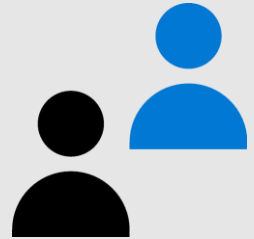


Private connectivity to services on Azure. Traffic remains on the Microsoft network, with no public internet access

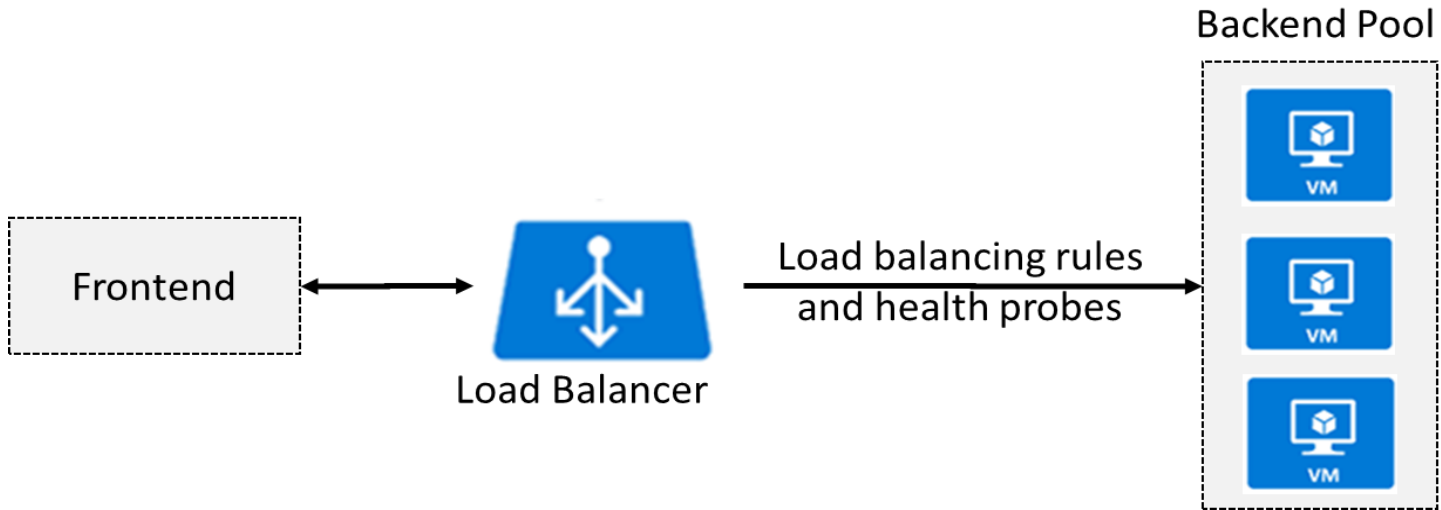
Integration with on-premises and peered networks

In the event of a security incident within your network, only the mapped resource would be accessible

Lesson 02: Configure Azure Load Balancer



Determine Azure Load Balancer Uses

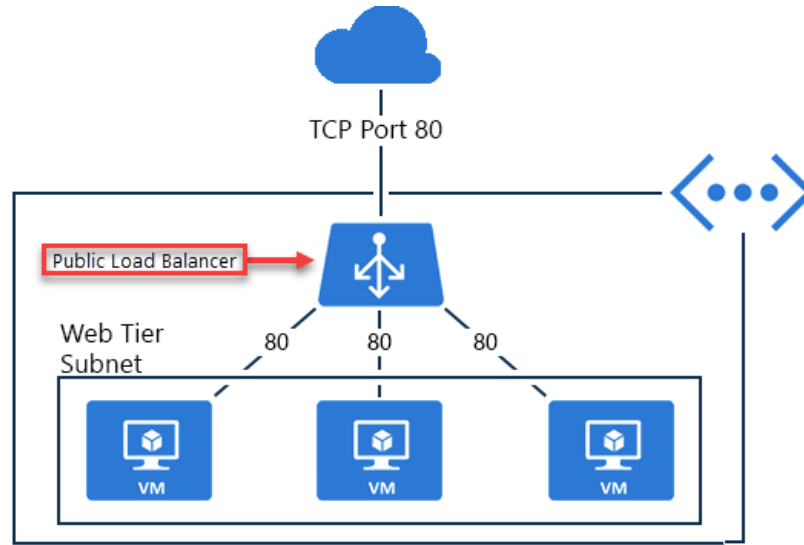


Distributes inbound traffic to backend resources using load-balancing rules and health probes

Can be used for both inbound/outbound scenarios

Two types: Public and Internal

Implement a Public Load Balancer



Maps public IP addresses and port number of incoming traffic to the VM's private IP address and port number, and vice versa

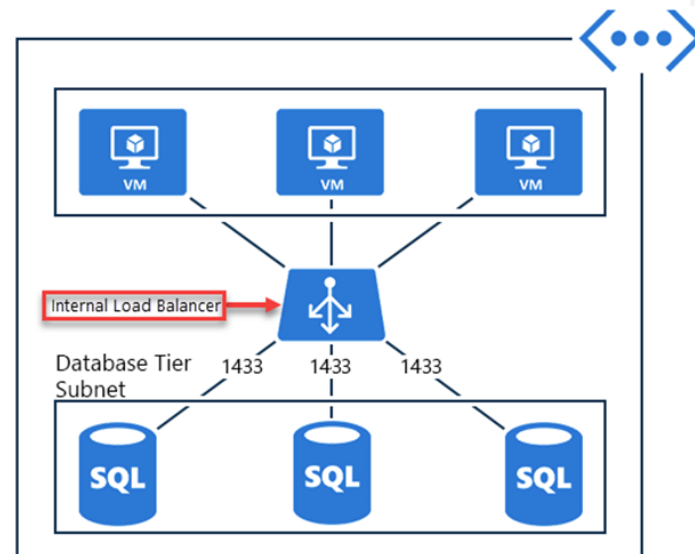
Apply load balancing rules to distribute traffic across VMs or services

Implement an Internal Load Balancer

Directs traffic only to resources inside a virtual network or that use a VPN to access Azure infrastructure

Frontend IP addresses and virtual networks are never directly exposed to an internet endpoint

Enables load balancing within a virtual network, for cross-premises virtual networks, for multi-tier applications, and for line-of-business applications



Determine Load Balancer SKUs

Feature	Basic SKU	Standard SKU
Backend pool	Up to 300 instances	Up to 1000 instances
Health probes	TCP, HTTP	TCP, HTTP, HTTPS
Availability zones	Not available	Zone-redundant and zonal frontends for inbound and outbound traffic
Multiple frontends	Inbound only	Inbound and outbound
Secure by default	Open by default. NSG optional.	Closed to inbound flows unless allowed by a NSG. Internal traffic from the virtual network to the internal load balancer is allowed.
SLA	Not available	99.99%

Instance details

Name *

lb01

Region *

(US) East US

Type * ⓘ



Internal



Public

SKU * ⓘ



Basic



Standard

Configure virtual network.

Virtual network * ⓘ

vnet01

Subnet *

subnet01 (10.1.0.0/24)

[Manage subnet configuration](#)

IP address assignment *



Static



Dynamic

Create Backend Pools

SETTINGS

 Backend pools

* Name

cesbackendpool

Associated to ⓘ

Unassociated ^

Unassociated

Availability set

Single virtual machine

Virtual machine scale set

SKU	Backend pool endpoints
Basic SKU	VMs in a single availability set or VM scale set
Standard SKU	Any VM in a single virtual network, including a blend of VMs, availability sets, and VM scale sets

To distribute traffic, a back-end address pool contains the IP addresses of the virtual NICs that are connected to the load balancer

Create Load Balancer Rules

Maps a frontend IP and port combination to a set of backend pool and port combination

Rules can be combined with NAT rules

A NAT rule is explicitly attached to a VM (or network interface) to complete the path to the target

Add load balancing rule

lbr01

Name *

lbr01

IP Version *

☒ IPv4 ☐ IPv6

Frontend IP address * ⓘ

10.1.0.4 (LoadBalancerFrontEnd)

Protocol

☒ TCP ☐ UDP

Port *

80

Backend port * ⓘ

80

Backend pool ⓘ

bep01

Health probe ⓘ

hp01 (HTTP:80)

Session persistence ⓘ

None

Idle timeout (minutes) ⓘ

Floating IP (direct server return) ⓘ

☒ Disabled ☐ Enabled

Configure Session Persistence

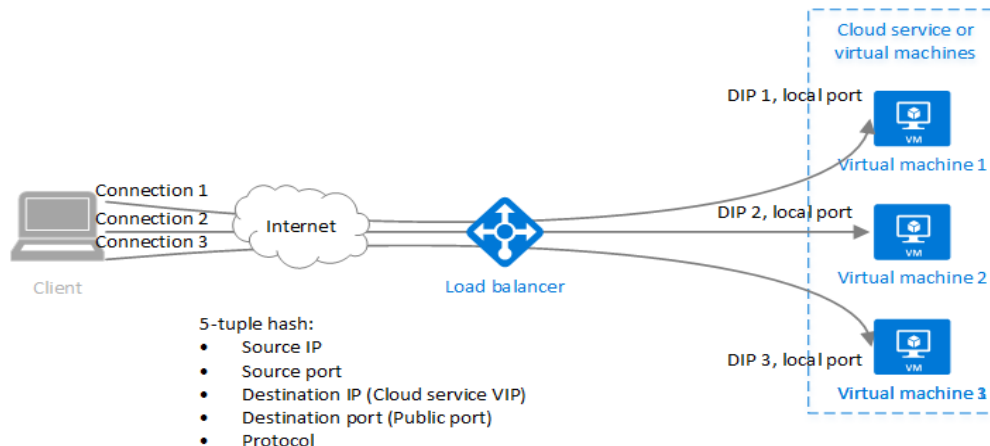
Session persistence ⓘ

None

None

Client IP

Client IP and protocol



Session persistence specifies how client traffic is handled

None (default) requests can be handled by any virtual machine

Client IP requests will be handled by the same virtual machine

Client IP and protocol specifies that successive requests from the same address and protocol will be handled by the same virtual machine

Create Health Probes

Allows the load balancer to monitor the status of an app

Dynamically adds or removes VMs from the load balancer rotation based on their response to health checks

HTTP custom probe (preferred) pings every 15 seconds

TCP custom probe tries to establish a successful TCP session

Add health probe

lb01

Name *

hp01

Protocol ⓘ

HTTP

Port * ⓘ

80

Path * ⓘ

/

Interval * ⓘ

5

seconds

Unhealthy threshold * ⓘ

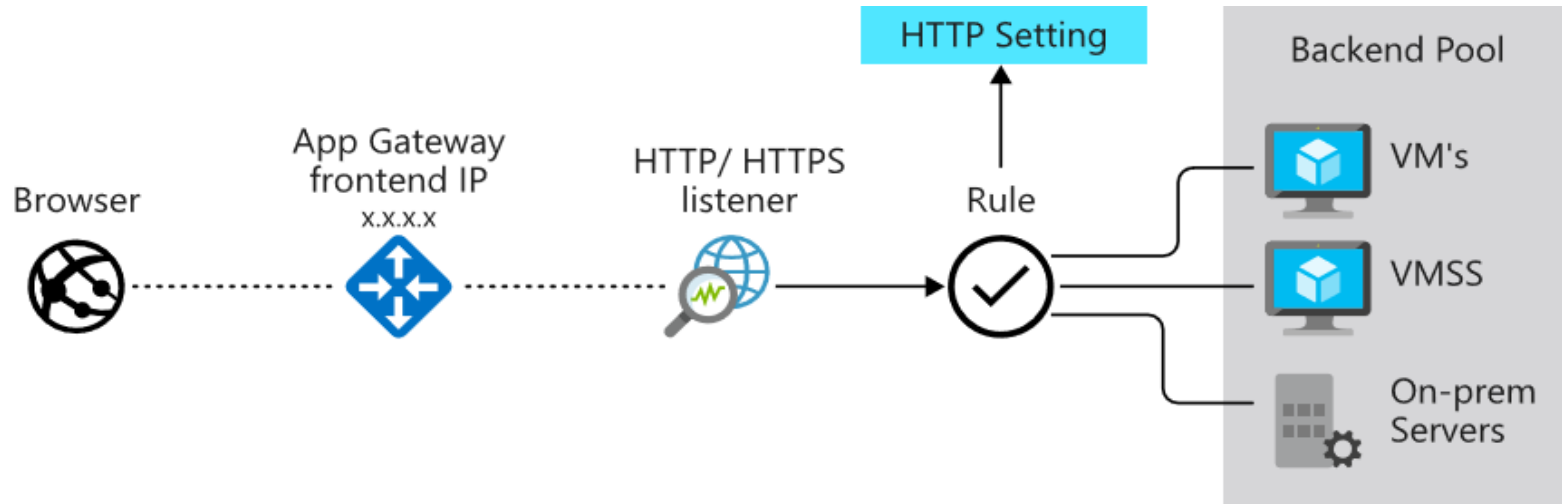
2

consecutive failures

Lesson 03 : Configure Azure Application Gateway



Implement Application Gateway



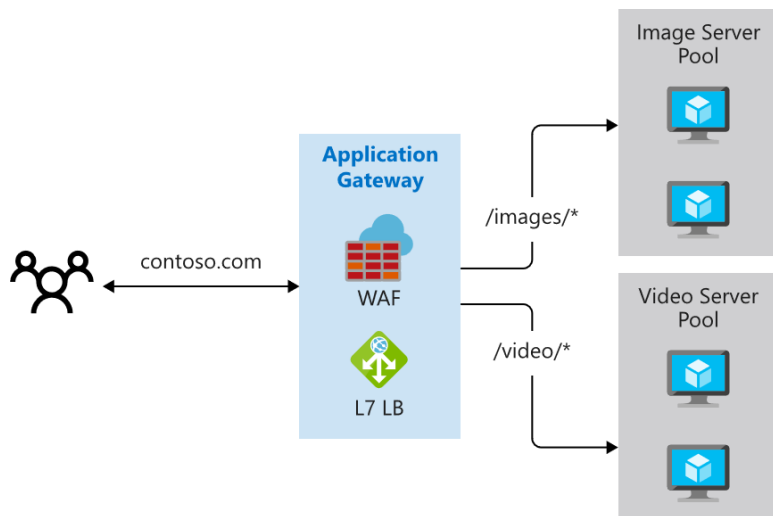
Manages web
app requests

Routes traffic to a pool of web
servers based on the URL of a
request

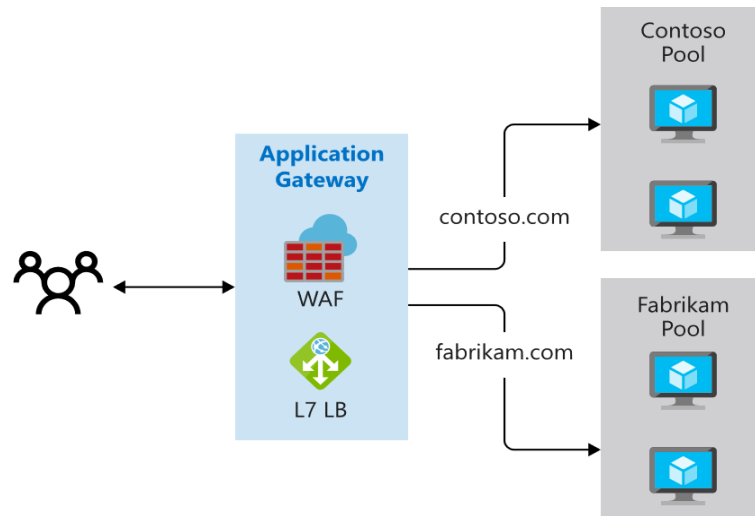
The web servers can be Azure virtual
machines, Azure virtual machine
scale sets, Azure App Service, and
even
on-premises servers

Determine Application Gateway Routing

Path-based routing



Multiple-site routing



Setup Application Gateway Components

Frontend IP

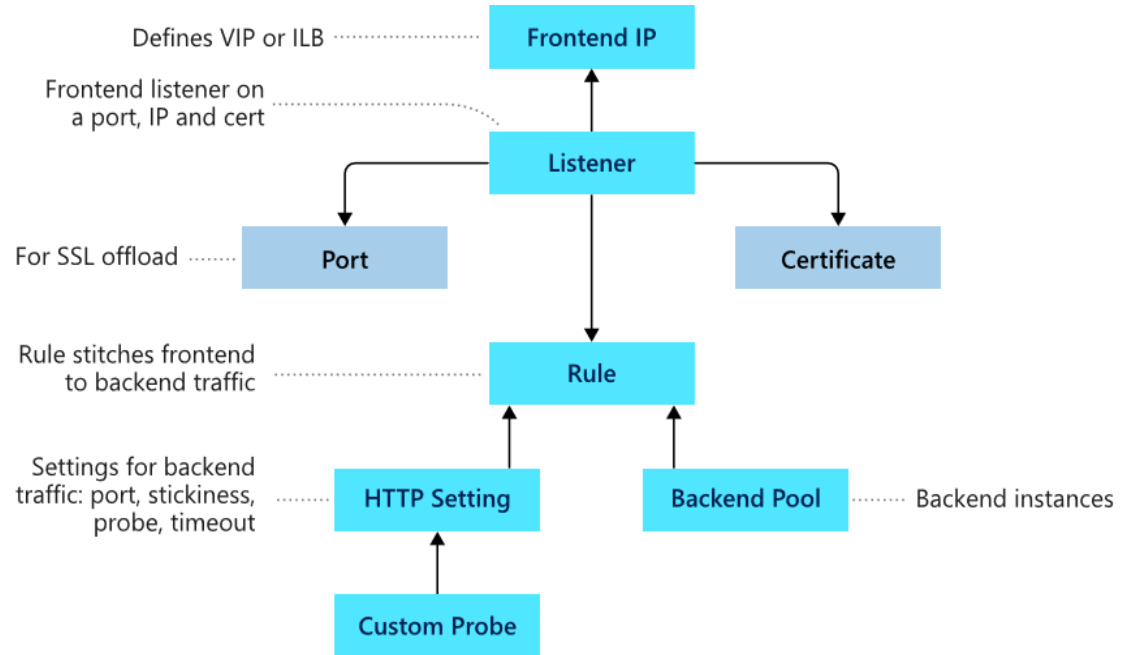
Listeners

Routing rules

Backend pools

Web application firewall
(optional)

Health probes



Thank you

